

CT4: CSE 201 - Object-Oriented Programming I: Java (Sec C)

Time: 20 min

Mark: 20

Name:

ID:

1. Create a user defined Exception "**InvalidContestantException**" which will take 2 int type parameter **min** and **max** and set the exception message to "You should be between **min** and **max** to participate in the contest.", here **min** and **max** are the parameters passed to the constructor. [6]
2. Define and implement a static method "**checkContestant**" which will take two parameters: a String "**name**" to take name of the contestant and an int type parameter "**age**" to take the age of the contestant. Inside the method, print the **name** and **age** parameter if the value of **age** is between 18 and 25. Otherwise, throw an **InvalidContestantException** and pass 18 and 25 for **min** and **max**. [8]
3. Now declare a main method. Inside the main method, take input for name and age of a contestant, call the **checkContestant** with these inputs. Take necessary step to handle the **InvalidContestantException** if age is not within the range. [6]

Note: Assume the **main** method and the **checkContestant** method are inside the same class.

CT4: CSE 201 - Object-Oriented Programming I: Java (Sec C)

Time: 20 min

Mark: 20

Name:

ID:

1. Create a user defined Exception "**InvalidStudentException**" which will take a String type parameter **id** and set the exception message to "**id** is not a valid student.", here **id** is the parameter passed to the constructor. [6]
2. Define and implement a static method "**checkStudent**" which will take two parameters: "**name**" and "**id**" of String type to take name and id of the student. Inside the method, print the **name** and **id** parameter if the **id** is a valid id. An id is valid if it is 8 character long and starts with "22201", throw an **InvalidContestantException** and pass 18 and 25 for **min** and **max**. [8]
3. Now declare a main method. Inside the main method, take input for name and age of a contestant, call the **checkContestant** with these inputs. Take necessary step to handle the **InvalidContestantException** if age is not within the range. [6]

Note: Assume the **main** method and the **checkContestant** method are inside the same class.

CT4: CSE 201 - Object-Oriented Programming I: Java (Sec D)

Time: 20 min

Mark: 20

Name:

ID:

1.a. Define and implement a static method “**checkVoter**” which will take an int type parameter “**age**”. Inside the method, print the **age** parameter if its value is 18 or greater. If **age** is less than 18, throw an **InvalidParameterException** with a message saying "Should be at least 18 years old to be eligible for vote." [8]

1.b. Now declare a main method. Inside the main method, call the **checkVoter** and pass an integer value taken as an input from user. Take necessary step to handle the **InvalidParameterException** if user input is less than 18. [6]

Note: Assume the **main** method and the **checkVoter** method are inside the same class.

2. Create a user defined Exception “**InvalidSelectionException**” which will take an int type parameter **input** and set the exception message to “**input** is not a valid selection”, here **input** is the parameter passed to the constructor. [6]

CT4: CSE 201 - Object-Oriented Programming I: Java (Sec D)

Time: 20 min

Mark: 20

Name:

ID:

- 1.a. Define and implement a **static** method “**checkAttendance**” which will take two int type parameter, “**attendance**” and “**totalClasses**”. Inside the method, calculate the percentage of attendance (using the value of **attendance** and **totalClasses**). If calculated percentage is less than 70, throw an **InvalidParameterException** with a message saying "Attendance percentage should be 70 or more to sit for the final exam". If calculated percentage is 70 or more, print "Eligible for final exam." [8]
- 1.b. Now declare a main method. Inside the main method, take input for **attendance** and **totalClasses** from user and call the **checkAttendance**. Take necessary step to handle the **InvalidParameterException** if **checkAttendance** throw the exception. [6]
- Note:** Assume the **main** method and the **checkAttendance** method are inside the same class.
2. Create a user defined Exception “**InvalidItemException**” which will take a String type parameter **input** and set the exception message to “**input** is not a valid item.”, here **input** is the parameter passed to the constructor. [6]

CT4: CSE 201 - Object-Oriented Programming I: Java (Sec E)

Time: 20 min

Mark: 20

Name:

ID:

1. Create a user defined Exception "**OverflowException**" which will take an int parameter **maxCapacity** and set the exception message to "Cannot hold more than **maxCapacity** guest in the centre.", here **maxCapacity** is the parameter passed to the constructor. [6]
2. Define and implement a static method "**reserve**" which will take two parameters: **eventName** of String type and **noOfGuests** of int type to represent event name and maximum no. of guests. Inside the method, print the "The resource has been reserved for **eventName** with capacity: **noOfGuests**" if **noOfGuests** is 500 or less. Otherwise, throw an **OverflowException** and pass 500 for **maxCapacity**. [8]
3. Now declare a main method. Inside the main method, take input for event name and total number of guests, call the **reserve** with these inputs. Take necessary step to handle the **OverflowException** if no of guests is not within the range. [6]

Note: Assume the **main** method and the **reserve** method are inside the same class.

CT4: CSE 201 - Object-Oriented Programming I: Java (Sec E)

Time: 20 min

Mark: 20

Name:

ID:

1. Create a user defined Exception "**InvalidContestantException**" which will take 2 int type parameter **min** and **max** and set the exception message to "You should be between **min** and **max** to participate in the contest.", here **min** and **max** are the parameters passed to the constructor. [6]
2. Define and implement a static method "**checkContestant**" which will take two parameters: a String "**name**" to take name of the contestant and an int type parameter "**age**" to take the age of the contestant. Inside the method, print the **name** and **age** parameter if the value of **age** is between 18 and 25. Otherwise, throw an **InvalidContestantException** and pass 18 and 25 for **min** and **max**. [8]
3. Now declare a main method. Inside the main method, take input for name and age of a contestant, call the **checkContestant** with these inputs. Take necessary step to handle the **InvalidContestantException** if age is not within the range. [6]

Note: Assume the **main** method and the **checkContestant** method are inside the same class.